

Aaron Dutta

647-244-5263 | Toronto, Ontario | aaron.dutta@uwaterloo.ca | [linkedin.com/in/aaron-dutta](https://www.linkedin.com/in/aaron-dutta) | github.com/defAaron | aarondutta.com

EDUCATION

University of Waterloo <i>BMath, Honours Mathematics with Co-Op</i>	Expected June 2031 <i>Waterloo, ON</i>
---	---

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, HTML5/CSS3, SQL
Libraries/Frameworks: FastAPI, Flask, Next.js, Prisma, React, Streamlit, Tailwind CSS
Developer Tools: Anthropic API, Docker, Figma, Gemini API, Git, MediaPipe, Microsoft Azure, NetworkX, NumPy, ONNX Runtime, OpenCV, Pandas, PostgreSQL, Pydantic, RESTful APIs, SQLite, Supabase, Vercel, YOLOv8

PROJECTS

TechniqueTitan <i>Python, MediaPipe, OpenCV, NumPy, FastAPI, Pandas, Matplotlib</i>   2026 - Built an AI piano posture coach that turns any device camera into on-demand technique feedback so students can correct hand form without a teacher - Engineered a MediaPipe + OpenCV + NumPy pipeline to extract 21 landmarks per hand, normalize for camera distance, and score 5 posture criteria - Shipped a FastAPI + React/TypeScript app with on-device browser MediaPipe and YAML-tunable thresholds, cutting review from a full lesson cycle to seconds - Built a batch CLI that walks image folders and exports CSV/JSON metrics along with a 45-test pytest suite across geometry, scoring, and API modules, automating dataset-scale analysis that was previously manual
SkyEye <i>Python, Flask, YOLOv8, ONNX Runtime, React, TypeScript, Gemini API</i>   2026 - Won RescueHacks 2026 for an SAR triage tool that turns a free-text missing-person report into a Lost Person Behavior search ring, scoring map pins in 5–8 s within a 30 s time-to-first-lead target - Developed a sliding-window YOLOv8n + ONNX Runtime pipeline to tile images into overlapping windows - Shipped a Flask + React/TypeScript operator console with Gemini Flash (Groq fallback) for typed extraction and Google Geocoding + LPB $\sqrt{t}$ radius
FraudGen <i>Python, Anthropic API, FastAPI, Streamlit, Pandas, NetworkX, Pydantic</i>   2026 - Designed a 4-agent adversarial LLM pipeline that explores fraud variant space via a coverage matrix and generates persona-seeded mule-network scenarios - Shipped async parallel orchestration, deterministic 8-rule validation/repair, and critic scoring with retry loops; a single produced variant labeled 11 transactions across a 3-victim → 4-mule converging funnel - Built a Streamlit console with network graph visualization and ML-ready exports for GNN training

CERTIFICATIONS

Essentials with Azure Fundamentals, Microsoft	2026
Google AI Essentials Specialization, Google	2026
Building with the Claude API, Anthropic	2026
CS50x, Harvard University	2024

AWARDS

Winner @ RescueHacks	2026
2nd Place Overall @ Katy Youth Hacks	2026
3rd Place Security/Privacy Hack @ STEMist Hacks	2026
1st Place @ STEMMINDS Hackathon	2026

EXPERIENCE

Mathematics Mentor <i>Filix School of Education</i>	2024 – Present <i>Purulia, India (Remote)</i>
- Mentoring 15+ Grade 7–12 students through weekly math circles, encouraging collaborative problem solving - Advancing students' critical thinking skills by combining theoretical concepts with practical exercises	